

ADA-Marker Answer Sheet

Anonymized local demo

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PROBLEM

Q1

Answer

3. (1) The possible biggest number s.t. $S[1] \dots S[z] = S[i] \dots S[i+z-1]$
- (2) We can establish an algorithm called 左右護法. $右[i]$ is defined as the biggest $(j + z(j) - 1)$, $1 \leq j \leq i$, $左[i]$ is defined as the j that satisfy $右[j] = i$
- (3) There are three cases: Case 1: $i \geq 右[i-1]$: Then it takes $z(i) + 1$ to find $z(i)$; Case 2: $i < 右[i-1]$ & $z(i) \leq 右[i-1] - i + 1$: It takes $O(1)$ to find $z(i)$; Case 3: $i < 右[i-1]$ & $z(i) > 右[i-1] - i + 1$: It takes $z(i) - 右[i-1] + i$ to find $z(i)$. Since All three cases take $O(z(i) + 1)$ time, This algorithm takes $O(n)$ time.
- (4) With the above saying, there are only three cases, and We have derived solutions for all the three cases. Hence, this algorithm will work.
- (5) The worst way to obtain a longest common substring is by comparing each m character for n times (supposed $n \geq m$), which takes $O(mn)$ time. Since the z -value algorithm takes $O(n)$, which is obviously a better algorithm. So the upper bound $O(mn)$ will also be the upper bound for z -value algorithm.